

1-0 TOOLS AND EQUIPMENT

SPECIAL TOOLS			
ITEM	PART #	DESCRIPTION	QTY
1	46-317048P1`	DecouplingTest Phantom	1
2	46-194427P49	Beckman 3030 DVM	1
3	46-255386P40	Vector Impedance Mater	1

FIELD SUPPLIED TOOLS			
ITEM		DESCRIPTION	QTY
1		15 mm open end wrench	1
2		5/16 inch open end wrench	1
3		7/16 inch open end wrench	1
4		3/32 inch transfer punch	1
5		3/16 inch flat blade screwdriver	1
6		3/4 inch open end wrench	1
7		#2 Phillips Screwdriver	1
8		3/8 inch open end wrench	1
9		Pliers	1
10		Small hammer	1
11		Utility knife	1
12		6 foot ruler	1

2-0 QUAD T/L COIL INSTALLATION INSTRUCTIONS

2-1 System Polarity

Before any quadrature coil can be used on the Signa® Advantage™ 0.5T System, the system polarity must be determined and the user coil adapted to reflect the same polarity.

2-1-1 Determining System Polarity

- Select [**NEW EXAM**] to enable the patient table **LANDMARK** function. Locate the surface coil decoupling phantom on the patient table. Do not place the Quad T/L Spine Coil under the decoupling phantom. Center the decoupling phantom; X, Y, Z using the positioning lights, **LANDMARK** and advance to isocenter. Set up a scan according to the parameters shown in ILLUSTRATION 2-1. At the **PREVIEW SCREEN**, press **SCAN OPS.** **DO NOT SCAN AT THIS TIME.**
- Press [**DISPLAY CVS**]. Press [**NEXT**] until the screen appears with **cfxfull** in the column under **CV Name**.
- Observe the **Current Value** column across from **cfxfull**. If there is a (-) sign in front of the number, the system polarity is **NEGATIVE**. If no sign appears in front of the number, the system polarity is **POSITIVE**.

2-1-2 Checking Coil Polarity

Remove the access cover shown in ILLUSTRATION 2-2 by removing the six nylon screws. Slide the Coil Positioning Handle until the balun board is visible in the access area. Ensure that the SMA connector is connected to the same polarity fitting as the system polarity. If not, remove the SMA connector from the fitting on the balun board, and connect it to the other fitting.

<u>PATIENT STUDY PARAMETERS</u>		<u>SCANNING RANGE</u>	
ID:	GESERVICE	Field of View:	[32]
Name:		Slice Thickness:	[5 mm]
Patient Weight:	111 LBS	Interscan Spacing:	[0]
	[PATIENT POSITION]	Start Location:	0
	[PATIENT POSITION]	End Location:	0
		No of Scan Locations	1
		FOV Center (L/R)	0 (P/A) 0
			[<] [ACQUISITION TIME]
<u>PATIENT POSITION</u>		<u>ACQUISITION TIME</u>	
Patient Entry:	[HEAD FIRST]	Acq. Matrix:	[256 x 128]
Patient Position:	[SUPINE]	Frequency Direction:	[R/L]
Axial/Sag Landmark:	[SYMPH PUBIS]	Imaging Time:	[1 Nex]
Coil Type:	[BODY]	Contrast:	[NO]
Scan Plane:	[AXIAL]	Table Delta	0
	[IMAGING PARAMETERS]		[SCAN SETUP]
<u>IMAGING PARAMETERS</u>		<u>SCAN SETUP</u>	
Image mode:	[2D]	Auto CF:	[WATER]
Pulse Sequence:	[SPIN ECHO]	Receive B/W (+/-):	[16 KHz]
Imaging Options:	[NONE]	Receive B/W (echos 2-4):	[16 KHz]
Enter PSD Filename:		Primary Arch. Node:	Default
Monitor SAR?:	Y		[REVIEW SCREEN]
	[NEXT SCREEN]		
	[SCAN TIMING]		
<u>SCAN TIMING</u>		<u>REVIEW</u>	
Number of Echos:	[4]		[SCAN OPS]
Echo Time (TE):	[20 ms]		
Rep Time (TR):	[OTHER] 600 ms	<u>SCAN OPERATIONS</u>	
	[SCANNING RANGE]		[AUTO PRESCAN]
			[SCAN]

SYSTEM TEST SCAN PARAMETERS

ILLUSTRATION 2-1

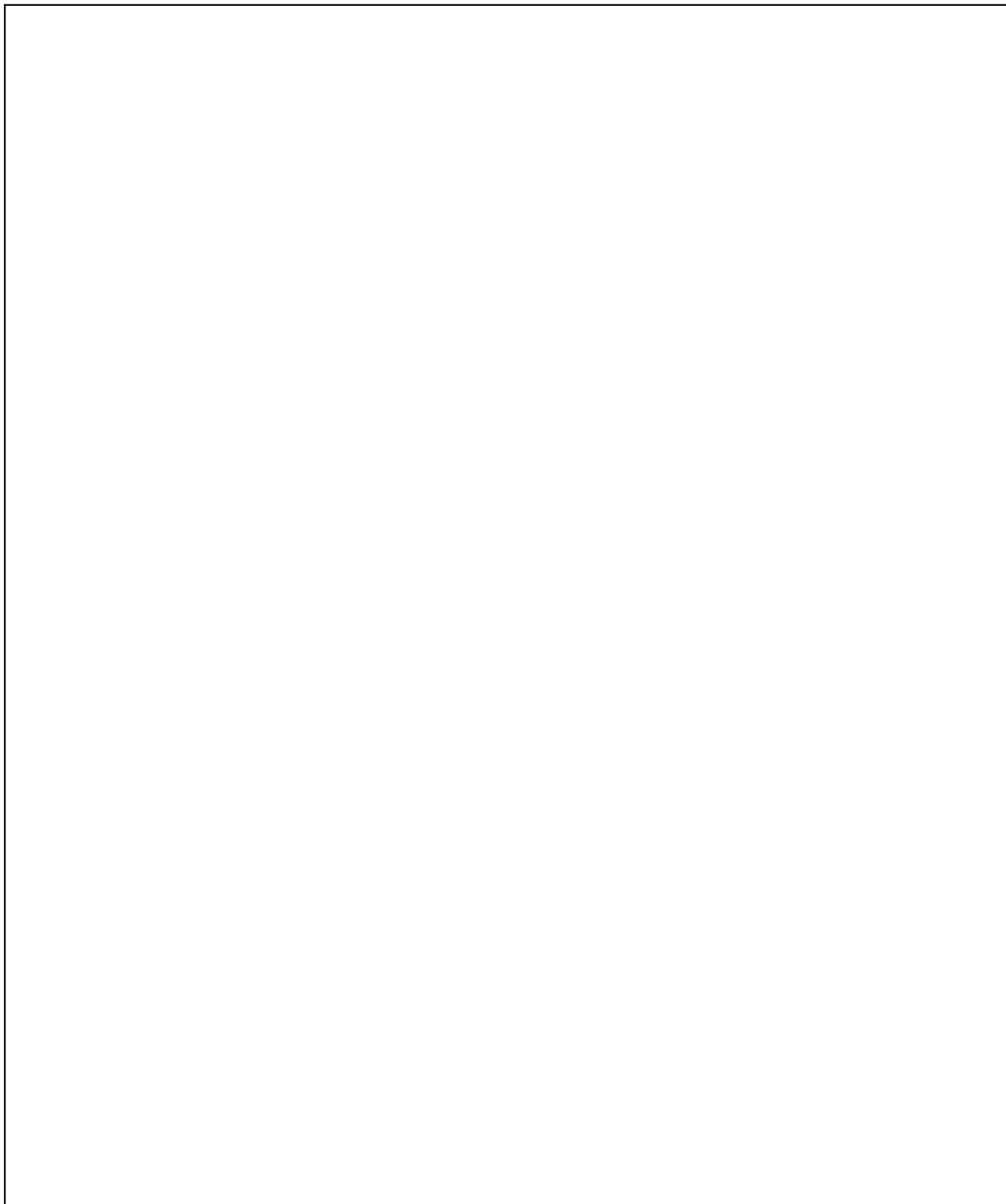
3-0 SIGNA ADVANTAGE 0.5T SCANNER VERIFICATION

- 3-1 Scanner performance in the body coil mode should be verified. The RF power level calibration from the body coil mode is used to verify the surface coil decoupling.

NOTE

The surface coil decoupling phantom must be filled with distilled water and the supplied cupric sulfate prior to use for any imaging tests. To properly prepare the phantom, completely fill the two liter Nalgene bottle supplied with the coil with distilled water while mixing in the vial of cupric sulfate. Ensure that all of the cupric sulfate has dissolved, and that there are no air bubbles when the cap is replaced on the bottle.

- 3-2 The parameters have been loaded in step 2-1-1. Perform an **AUTO PRESCAN** to properly calibrate the RF power level. Record the value for TG.
- 3-3 Run the scan. Observe the resulting images. Ensure that there are no artifacts of any sort on any of the images.
- 3-4 Select [**CANCEL**] and [**NEW SERIES**] to perform SNR comparisons. Acquire two identical images using the scan parameters in ILLUSTRATION 3-2. After acquiring the first image, select **SCAN** to obtain a second. Note the series and image numbers for future reference.



COIL POLARITY CONNECTION, TEST POINTS FOR DIODE CHECKS

ILLUSTRATION 2-2

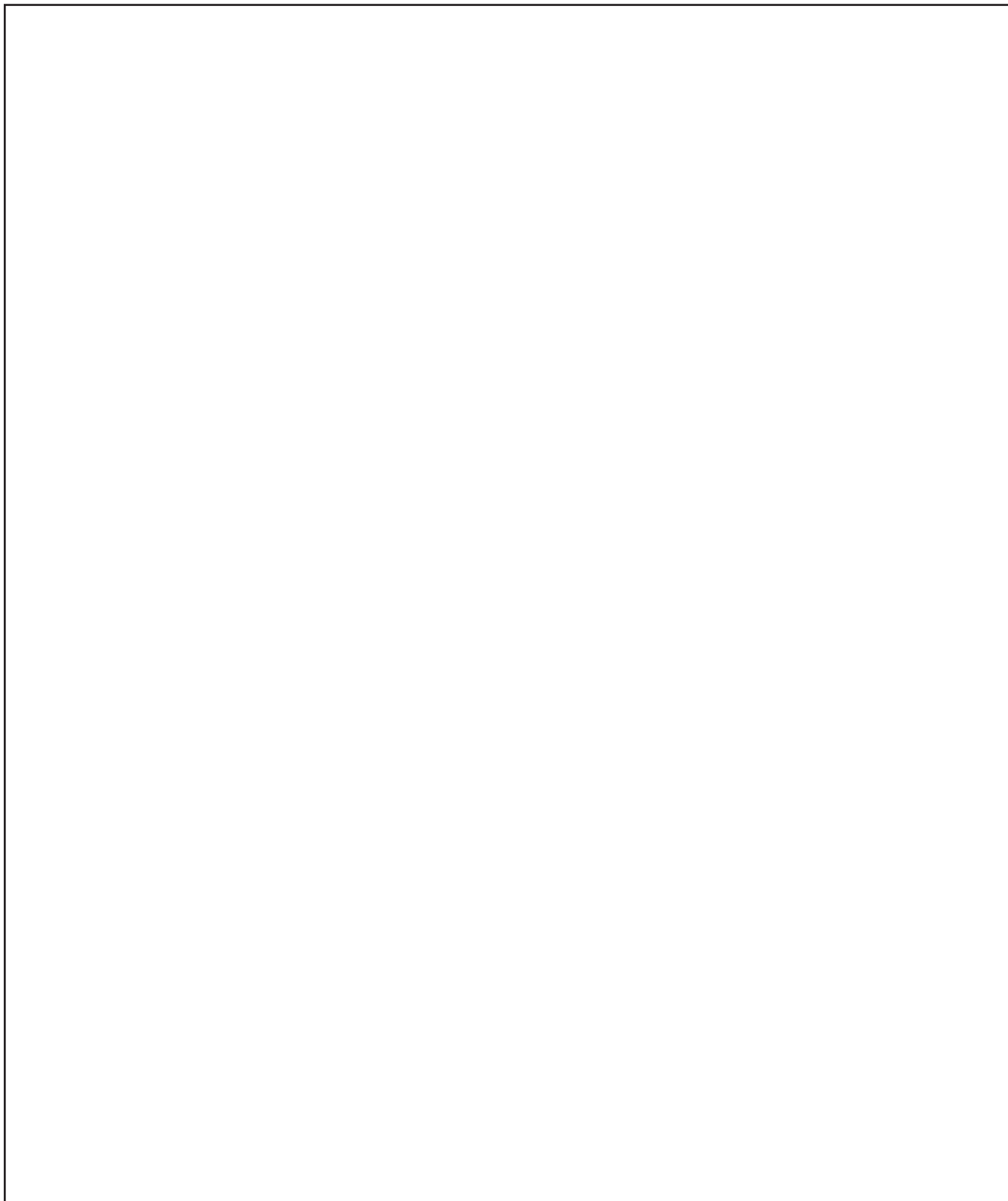
PATIENT STUDY PARAMETERS		SCANNING RANGE	
ID:	GESERVICE	Field of View:	[32]
Name:		Slice Thickness:	[5 mm]
Patient Weight:	111 LBS	Interscan Spacing:	[0]
	[PATIENT POSITION]	Start Location:	0
		End Location:	0
		No of Scan Locations	1
		FOV Center (L/R)	0 (P/A) 0
			[<] [ACQUISITION TIME]
PATIENT POSITION		ACQUISITION TIME	
Patient Entry:	[FEET FIRST]	Acq. Matrix:	[256 x 128]
Patient Position:	[SUPINE]	Frequency Direction:	[R/L]
Axial/Sag Landmark:	[SYMPH PUBIS]	Imaging Time:	[1 Nex]
Coil Type:	[BODY]	Contrast:	[NO]
Scan Plane:	[SAGITTAL]	Table Delta	0
	[IMAGING PARAMETERS]		[SCAN SETUP]
IMAGING PARAMETERS		SCAN SETUP	
Image mode:	[2D]	Auto CF:	[WATER]
Pulse Sequence:	[SPIN ECHO]	Receive B/W (+/-):	[16 KHz]
Imaging Options:	[NONE]	Primary Arch. Node:	Default
Enter PSD Filename:			[REVIEW SCREEN]
Monitor SAR?:	Y		
	[NEXT SCREEN]		
	[SCAN TIMING]	REVIEW	[SCAN OPS]
SCAN TIMING		SCAN OPERATIONS	
Number of Echos:	[1]		
Echo Time (TE):	[20 ms]		
Rep Time (TR):	[OTHER] 600 ms		[AUTO PRESCAN]
	[SCANNING RANGE]		[SCAN]

BODY COIL SNR SCAN PARAMETERS

ILLUSTRATION 3-2

4-0 QUAD T/L SPINE COIL IMAGING PERFORMANCE VERIFICATION

- 4-1 Set up the surface coil and decoupling phantom on the patient table as described in ILLUSTRATION 4-1. Center the decoupling phantom; X, Y, Z using the positioning lights, **LANDMARK** and advance to isocenter.
- 4-2 Select **[CANCEL]** and **[NEW SERIES]** to set up the scan using the parameters in ILLUSTRATION 4-2. Use **AUTO PRESCAN** to set the values of R1 and R2 to those of Section 3. If the value of TG is different from that obtained in step 3-3, use **MANUAL PRESCAN** to set the value of TG to the value found in step 3-3.
- 4-3 Run the scan. Ensure that there are no artifacts of any sort on any of the images. Study the third and fourth echoes using a **WINDOW WIDTH** and **WINDOW LEVEL** that provide good image contrast. A properly functioning coil will produce an image with a smooth signal pattern, gradually dropping in intensity as the distance from the coil increases, as shown in ILLUSTRATION 4-3. A decoupling failure will cause holes, distortions, or striped patterns in the image as seen in ILLUSTRATION 4-4. Refer to Section 6-0 to troubleshoot decoupling failures.
- 4-4 Select **[CANCEL]** and **[NEW SERIES]** to acquire two identical images using the scan parameters in ILLUSTRATION 4-5. After acquiring the first image, select **SCAN** to obtain a second.
- 4-5 Compute the SNR using the procedure described in Section 5-0 '**SNR IMAGE ANALYSIS**'. Compare the signal to noise ratio of the first echo of the Quad T/L Spine Coil images with the signal to noise ratio of the first echo of the body coil images of step 3-4. The Quad T/L Spine Coil scans should display a signal to noise ratio greater than 2.5 times that of the body coil scans.



DECOUPLING PHANTOM PLACEMENT

ILLUSTRATION 4-1

<u>PATIENT STUDY PARAMETERS</u>		<u>SCANNING RANGE</u>	
ID:	GESERVICE	Field of View:	[32]
Name:		Slice Thickness:	[5 mm]
Patient Weight:	111 LBS	Interscan Spacing:	[0]
	[PATIENT POSITION]	Start Location:	0
		End Location:	0
		No of Scan Locations	1
		FOV Center (L/R)	0 (P/A) 0
			[<] [ACQUISITION TIME]
<u>PATIENT POSITION</u>		<u>ACQUISITION TIME</u>	
Patient Entry:	[HEAD FIRST]	Acq. Matrix:	[256 x 128]
Patient Position:	[SUPINE]	Frequency Direction:	[R/L]
Axial/Sag Landmark:	[SYMPH PUBIS]	Imaging Time:	[1 Nex]
Coil Type:	[OTHER] [T/L SPINE]	Contrast:	[NO]
Scan Plane:	[AXIAL]	Table Delta	0
	[IMAGING PARAMETERS]		[SCAN SETUP]
<u>IMAGING PARAMETERS</u>		<u>SCAN SETUP</u>	
Image mode:	[2D]	Auto CF:	[WATER]
Pulse Sequence:	[SPIN ECHO]	Receive B/W (+/-):	[16 KHz]
Imaging Options:	[NONE]	Receive B/W (echos 2-4):	[16 KHz]
Enter PSD Filename:		Primary Arch. Node:	Default
Monitor SAR?:	Y		[REVIEW SCREEN]
	[NEXT SCREEN]		
	[SCAN TIMING]		
<u>SCAN TIMING</u>		<u>REVIEW</u>	
Number of Echos:	[4]		[SCAN OPS]
Echo Time (TE):	[20 ms]		
Rep Time (TR):	[OTHER] 600 ms		
	[SCANNING RANGE]		
		<u>SCAN OPERATIONS</u>	
			[AUTO PRESCAN]
			[SCAN]

SURFACE COIL DECOUPLING TEST SCAN PARAMETERS

ILLUSTRATION 4-2



NORMAL DECOUPLING

ILLUSTRATION 4-3



DECOUPLING FAILURE

ILLUSTRATION 4-4

<u>PATIENT STUDY PARAMETERS</u>		<u>SCANNING RANGE</u>	
ID:	GESERVICE	Field of View:	[32]
Name:		Slice Thickness:	[5 mm]
Patient Weight:	111 LBS	Interscan Spacing:	[0]
	[PATIENT POSITION]	Start Location:	0
		End Location:	0
		No of Scan Locations:	1
		FOV Center (L/R):	0 [P/A] 0
<u>PATIENT POSITION</u>			[ACQUISITION TIME]
Patient Entry:	[FEET FIRST]	<u>ACQUISITION TIME</u>	
Patient Position:	[SUPINE]	Acq. Matrix:	[256 x 128]
Axial/Sag Landmark:	[SYMPH PUBIS]	Frequency Direction:	[R/L]
Coil Type:	[OTHER] [T/L SPINE]	Imaging Time:	[1 Nex]
Scan Plane:	[SAGITTAL]	Contrast:	[NO]
	[IMAGING PARAMETERS]	Table Delta:	0
<u>IMAGING PARAMETERS</u>			[SCAN SETUP]
Image mode:	[2D]	<u>SCAN SETUP</u>	
Pulse Sequence:	[SPIN ECHO]	Auto CF:	[WATER]
Imaging Options:	[NONE]	Receive B/W (+/-):	[16 KHz]
Enter PSD Filename:		Primary Arch. Node:	Default
Monitor SAR?:	Y		[REVIEW SCREEN]
	[NEXT SCREEN]		
	[SCAN TIMING]		
<u>SCAN TIMING</u>		<u>REVIEW</u>	
Number of Echos:	[1]		[SCAN OPS]
Echo Time (TE):	[20 ms]	<u>SCAN OPERATIONS</u>	
Rep Time (TR):	[OTHER] 600 ms		[AUTO PRESCAN]
	[SCANNING RANGE]		[SCAN]

SURFACE COIL SNR SCAN PARAMETERS

ILLUSTRATION 4-5

- 4-6 Inspect the images for visible ghosting [similar in appearance to motion artifacts] in the phase encode direction. If images contain ghosting artifacts, suspect an intermittent cable or connection. Refer to Section 6-0 for troubleshooting information.

5-0 SNR IMAGE ANALYSIS

Description

NEMA STANDARD SNR TEST

Filter
Off

Image
Off

First
Image

List/
Sel
Exams

List/
Sel
Series

List/
Sel
Images

Second
Image

List/
Sel
Series

List/
Sel
Images

Body Axial SNR Data

Signal : xx.x

Noise : xx.x

SNR : xx.x

Cancel

Utility
Main

MR
Tools
Main

Image
Qual-
ity

Accept



Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN
ILLUSTRATION 5-1

Coil	Signal to Noise Ratio	Date
Body Coil		
T/L Spine Coil		

SNR VALUES
ILLUSTRATION 5-2

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center of the image for positioning the ROI. A difference image is created by subtracting the second image from the first and the same ROI is used to calculate noise from the subtracted image. The signal value, noise value, and signal to noise ratio are reported. There is an option to save the difference image with the results annotated.

5-1 Procedure

5-1-1 Touch **[UTILITIES]**, **[MRTools]**, **[Image Quality]**, then **[SNRTest]**. The SNR Test screen is displayed. See ILLUSTRATION 5-1.

5-1-2 Enter first image exam, series, and image number. If exam, series, or image numbers are not known, select **[List Exams]**, **[List Series]**, or **[List Images]** to display list to choose from.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.

5-1-3 Enter second image series and image number. If series or image numbers are not known, select **[List Series]** or **[List Images]** to display list. The second image Exam Number is defaulted to the same as first image.

5-1-4 Select **[Filter Off]** and **[Image Off]**.

5-1-5 Touch **[Accept]** to begin analysis. The final values are displayed on the screen. See ILLUSTRATION 5-1.

5-1-6 Record Signal to Noise Ratio for future reference on ILLUSTRATION 5-2

5-1-7 Touch **[Continue]** then select the next exam and repeat the above analysis for each image pair.

6-0 TEST AND TROUBLESHOOTING PROCEDURES

6-1 Coil Impedance Measurements

6-1-1 Detach the patient table and remove it from the scan room to an area where the magnetic field from the SIGNA ADVANTAGE 0.5T magnet is less than 5 Gauss. Position the surface coil on the patient table. Load the surface coil with a human subject. Follow the procedure in the OPERATION MANUAL to properly locate the patient and surface coil.

6-1-2 Connect the surface coil cable to the vector impedance meter probe using an appropriate adapter. Use the vector impedance meter to measure the coil impedance at 21.29 MHz. Ensure that the cable does not touch or pass close to the coil housing while making this measurement.

- 6-1-3 The impedance magnitude should nominally be 50 ohms under loaded conditions. It should fall within the range of 37.5 to 62.5 ohms under test conditions.
- 6-1-4 The phase angle should nominally be zero degrees. It should fall within the range of -17 to +17 degrees under test conditions.

NOTE

Quadrature surface coils can not be tuned in the field. If the coil falls outside of the measurement limits in steps 6-1-3 and 6-1-4, signal to noise ratio may be degraded. Proceed
w i t h
Section 6-2.

6-2 Coil Integrity Tests

- 6-2-1 Repeat the steps in Section 6-1 while flexing the surface coil cable, especially at the fitting and the coil housing strain relief. Note any erratic fluctuations in the coil impedance using the **HIGH SPEED** mode on the vector impedance meter. Any erratic or unstable readings may indicate a defective cable or connector. Refer to Section 7-0 for service procedures.
- 6-2-2 Repeat the steps in Section 6-1 while sliding the coil over the range of travel in the positioner. Using the **HIGH SPEED** mode on the vector impedance meter, check for any erratic fluctuations in the coil impedance. If erratic readings are obtained, the internal coaxial cable or the coil assembly may be defective. Refer to Section 7-0 for service procedures.

6-3 Protection Diode Tests

- 6-3-1 Use the digital multimeter on the diode test range to measure across the coil connector. With the negative lead to the connector shell and the positive to the connector center pin, the reading should be between 0.5 and 1.0. Reversing the leads should cause the reading to be infinity [overrange].
- 6-3-2 An open reading in both directions indicates two open diodes, or an open cable. Refer to Section 6-5 to check the cables. Refer to Section 7-8 to replace protection diodes.
- 6-3-3 A short circuit or excessively low reading indicates one or more shorted PIN diodes or a shorted cable. Refer to Section 6-5 to check the cables. Refer to Section 7-8 to replace protection diodes.

6-4 Decoupling Diode Tests**WARNING**

Do not move the balun board components when making these measurements. Any movement of the components may detune the coil circuitry.

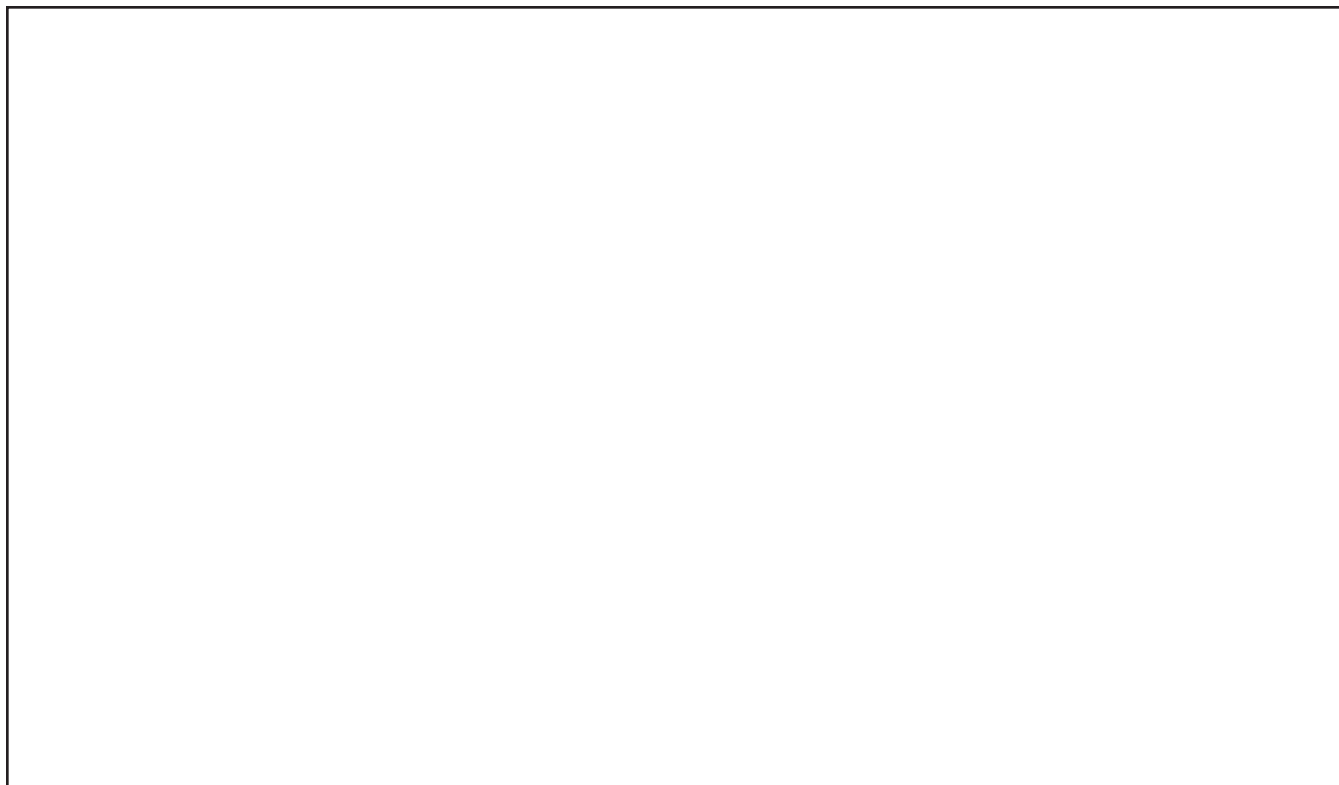
NOTE

Decoupling diodes are not field replaceable. Coils with decoupling diode failures must be replaced with a new coil assembly.

- 6-4-1 Remove the service access cover by carefully removing the six nylon screws as shown in ILLUSTRATION 2-2.
- 6-4-2 Locate the three pairs of test points for the decoupling diodes labeled V, H1, and H2 in the illustration detail. The coil itself is not marked.
- 6-4-3 Using the diodes test function on the digital multimeter, measure across each pair of points in sequence, first across the pair marked V. A reading between 0.5 and 1.0 should be obtained. Reverse the leads on the same test points to reverse the polarity of the multimeter. A similar reading should be obtained.
- 6-4-4 An overrange in both directions indicates that both diodes of a pair are open. This condition causes a portion of the decoupling circuitry to be inoperative. The coil should be replaced.
- 6-4-5 A reading near zero indicates one or more shorted diodes. While decoupling is not affected, the signal to noise ratio of the coil will be degraded, and the coil should be replaced.

6-5 Cable Tests

- 6-5-1 If the surface coil passes the tests in Sections 6-1, 6-2, and 6-3, the cables are assumed to be functional.



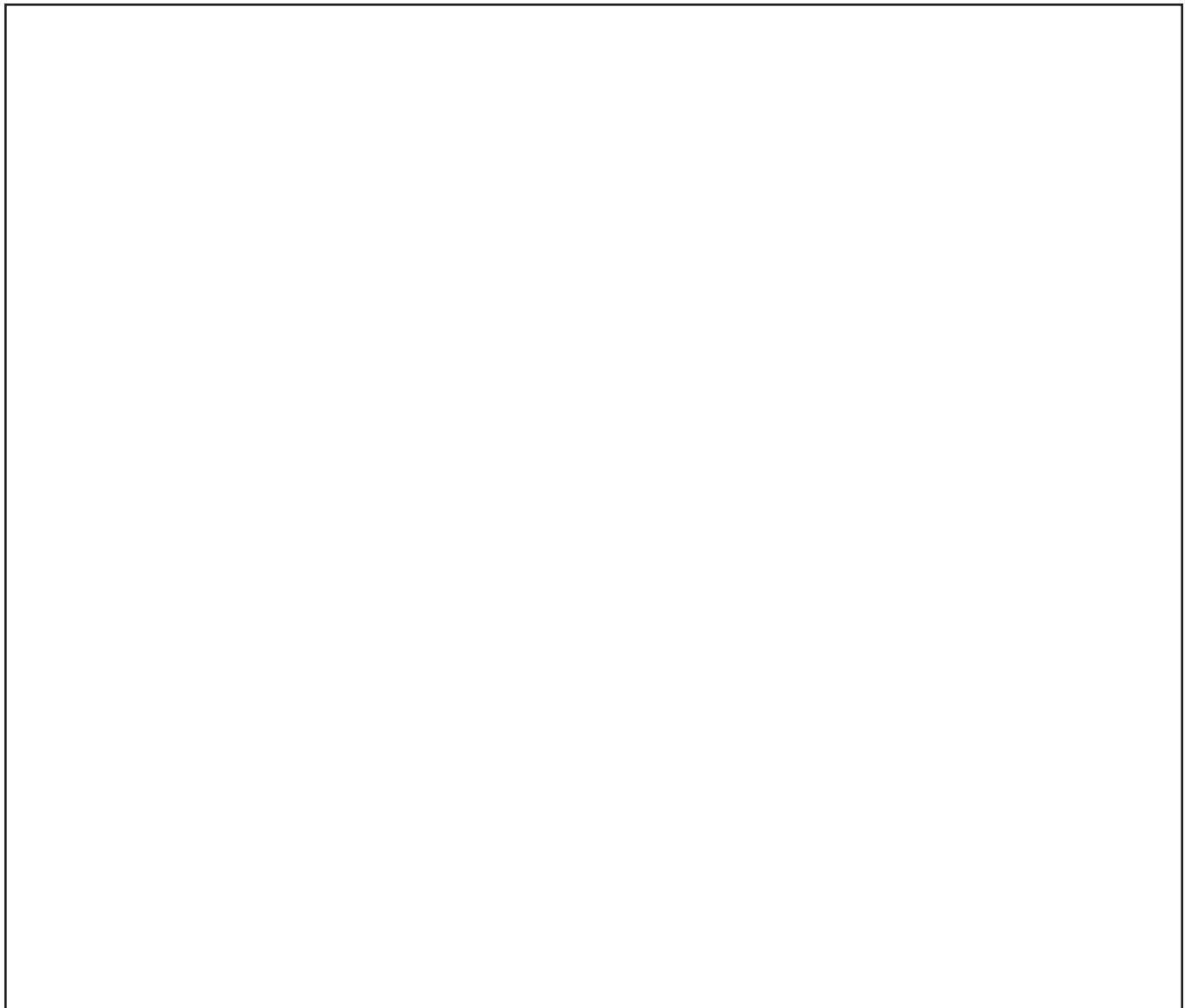
EXTERNAL COAXIAL CABLE REPLACEMENT

ILLUSTRATION 6-1

- 6-5-2 If a short or open circuit was discovered in Section 6-3, the cable may be tested by disconnecting the cable at the balun board via the service access cover, as shown in ILLUSTRATION 2-2. The cable may then be tested for continuity of the center conductor and shield, and for shorts between the center conductor and shield using the multimeter continuity test function.
- 6-5-3 If the cable system is found to be defective in step 6-5-2, the failed cable may be isolated by opening the junction between the internal and external cables at the SMA connector beneath the coil housing near the strain relief, and testing each cable separately. See ILLUSTRATION 6-1.

6-6 Verification of a Drive Cord Problem — ILLUSTRATION 6-2

- 6-6-1 Turn the coil housing over, allowing it to rest on its top cover.



POSITIONER ASSEMBLY/DISASSEMBLY

ILLUSTRATION 6-2

- 6-6-2 Slide the positioning handle to the end opposite the cable exit.
- 6-6-3 While looking through the clear service access cover, slide the handle back and forth and verify that the handle position is within +/- 0.25 inches of the actual coil position.
- 6-6-4 If the handle moves more than +/- 0.25 inch prior to coil movement in either direction, one or both of the drive cords have stretched beyond acceptable limits. If the coil does not move at all, one or both of the drive cords have been broken. Refer to Section 7-2 for repair procedures; refer to Section 7-5 for drive cord replacement and adjustment.

7-0 SERVICE PROCEDURES

7-1 Service Kits

- 7.1.1 A service kit is available for replacement of the external cable assembly. ILLUSTRATION 7-1 describes the contents of the kit.

CABLE KIT 46-317966P1 — 30523C27004 - Consisting of the following items:

ITEM	VENDOR PART#	DESCRIPTION	QTY
1	30511C27003	External Cable Assembly	1
2	87305-T-167	Service Manual - SIGNA 0.5T	1

EXTERNAL CABLE SERVICE KIT

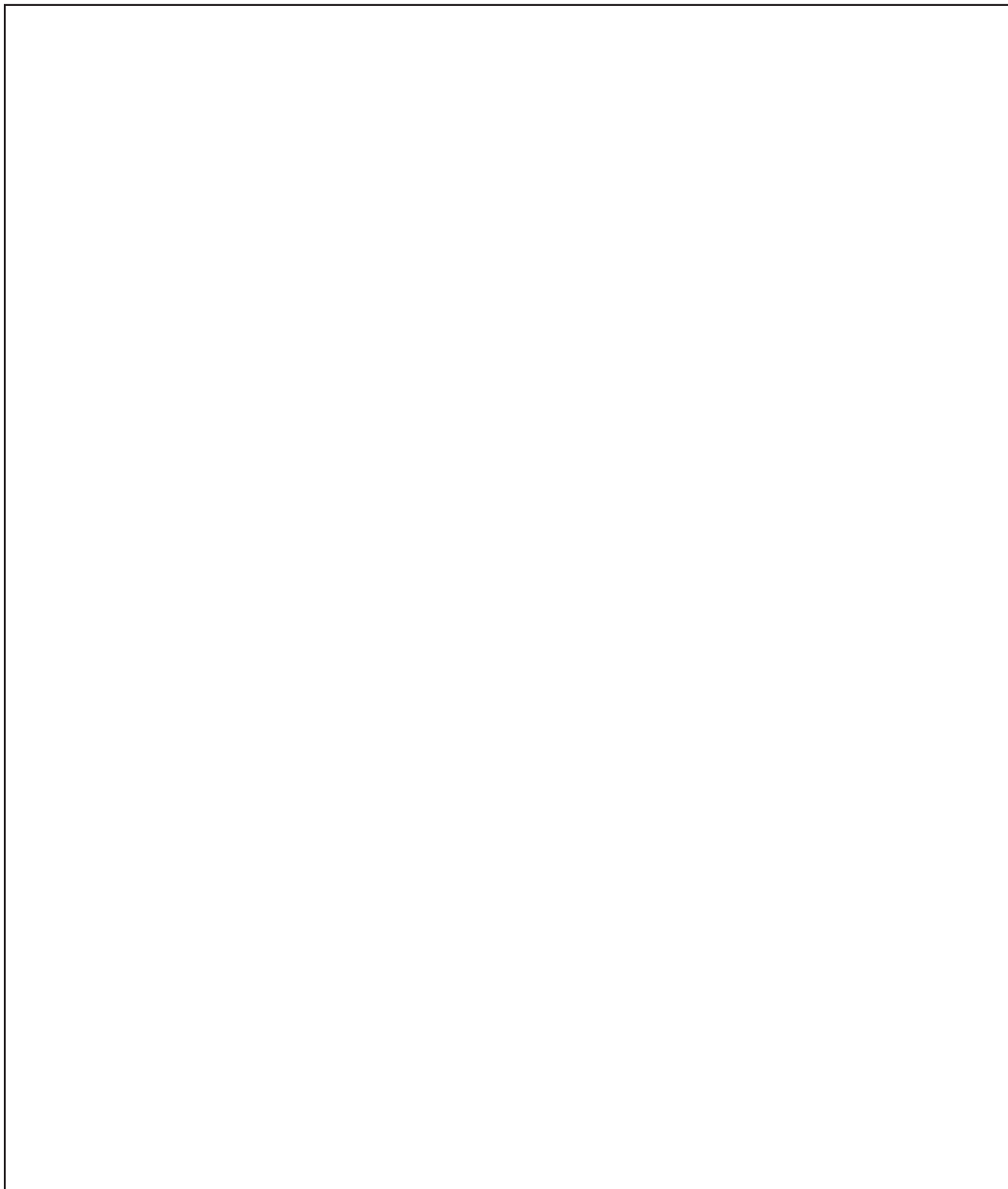
ILLUSTRATION 7-1

CABLE KIT 46-317047P1 - Consisting of the following items:

ITEM	VENDOR PART#	DESCRIPTION	QTY
1	708-0003-009	Drive Cord	12 ft
2	616-0218-450	Drive Rivets	9
3	670-0140-140	Binding Post Fastener	2
4	651-0440-381	4-40 Nylon Fastener	2
5	651-0150-260	Black Indicator Cap	2
6	30511C17002	Internal Cable Assembly	1
7	30519B15001	Clip-Audio Cable - MR MAX 0.5T only	2
8	87305-T-140	Service Manual - Signa 1.5T	1
9	87305-T-147	Service Manual - MR MAX 0.5T	1
10	87305-T-167	Service Manual - Signa 0.5T	1

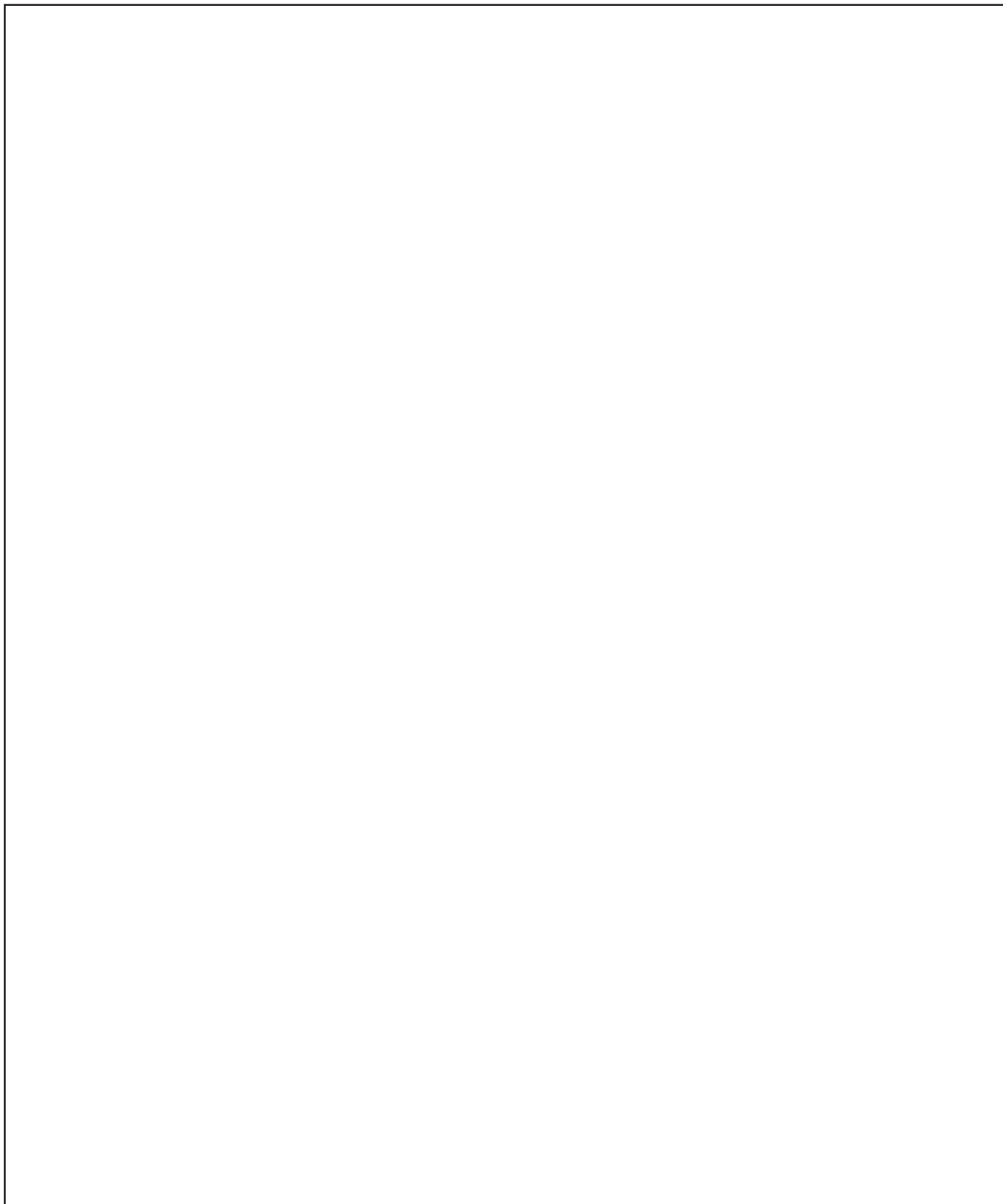
INTERNAL CABLE SERVICE KIT

ILLUSTRATION 7-2



EXTERNAL COAXIAL CABLE REPLACEMENT

ILLUSTRATION 7-3



POSITIONER ASSEMBLY/DISASSEMBLY

ILLUSTRATION 7-4

- 7-1-2 A service kit is available for internal service to the surface coil including the internal coaxial cable and the drive cords. ILLUSTRATION 7-2 describes the contents of the kit.

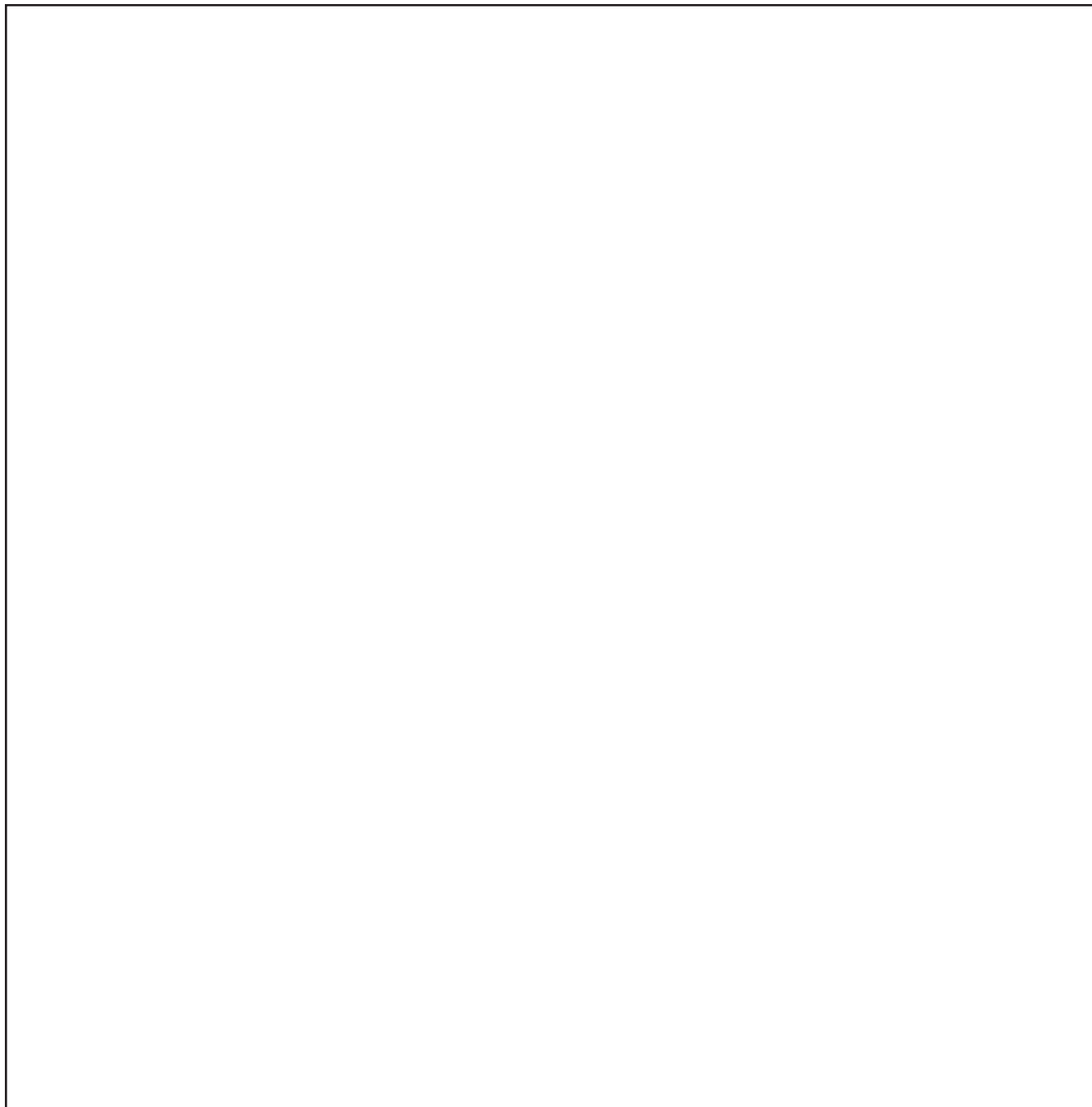
7-2 External Coaxial Cable Replacement ILLUSTRATION 7-3

- 7-2-1 Loosen and remove the strain relief body from the strain relief base using a 15 MM open end wrench. Slide the strain relief body down the cable away from the base.
- 7-2-2 Loosen and remove the retaining nut from the strain relief base using a 3/4 inch open end wrench.
- 7-2-3 Remove the cable SMA connector from the SMA feedthrough connector on the bottom cover of the housing.
- 7-2-4 Remove the strain relief base from the threaded hole in the top cover of the positioner using a 15 MM wrench.
- 7-2-5 Loosen and remove the new strain relief body from the new strain relief base using a 15 MM open end wrench. Remove the retaining nut from the new external cable assembly. Insert the SMA end of the cable through the strain relief mounting hole in the positioner top cover. Place the retaining nut over the SMA end of the cable.
- 7-2-6 Connect the cable SMA connector to the feedthrough connector on the positioner bottom cover. Tighten firmly
- 7-2-7 Attach the strain relief base to the top cover of the positioner housing. Tighten one-quarter turn beyond initial contact. Install and tighten the retaining nut using a 3/4 inch wrench. Tighten the nut one-half turn beyond initial contact with the surface of the top cover.
- 7-2-8 Using a 15 MM wrench, install and tighten the strain relief body until the base begins to turn [8 inch-pounds].
- 7-2-9 Repeat the tests in Sections 3-0, 4-0, and 5-0 to verify proper coil operation.

7-3 Positioner Disassembly Instructions - ILLUSTRATION 7-4

- 7-3-1 Remove the comfort pad from the top of the coil housing; turn the coil housing over to expose the bottom side.
- 7-3-2 Remove the clear service access cover by removing the six nylon screws.
- 7-3-3 Disconnect the internal cable from the balun board by unscrewing the SMA connector. Note the polarity setting for assembly of the coil.
- 7-3-4 Remove the external cable assembly as described in Section 7-2.
- 7-3-5 Remove the nine nylon screws located beneath the coil housing on the positioning handle side of the coil housing.

- 7-3-6 Remove the carrying handle by removing the two knurled nuts which retain the handle, and the two springs.
- 7-3-7 Remove the nine nylon drive rivets [two on each end, five on the carrying handle side of the coil housing] by punching the pin into the coil housing using a 3/32 inch transfer punch and a small hammer. Pull the rivet body out of the hole after the pin has been driven into the coil housing. See top of ILLUSTRATION 7-4.



INTERNAL COAXIAL CABLE REPLACEMENT

ILLUSTRATION 7-5

- 7-3-8 Remove the ten brass nuts and washers located on the bottom cover of the coil housing using an 11/32 nut driver.
- 7-3-9 Rest the coil housing on its bottom cover, and move the coil to the center of its travel. Carefully spread the slot sufficiently to push the positioning handle through the slot into the coil housing. Lift off the top cover.
- 7-3-10 Locate and remove the four drive rivet pins previously pushed into the coil housing.

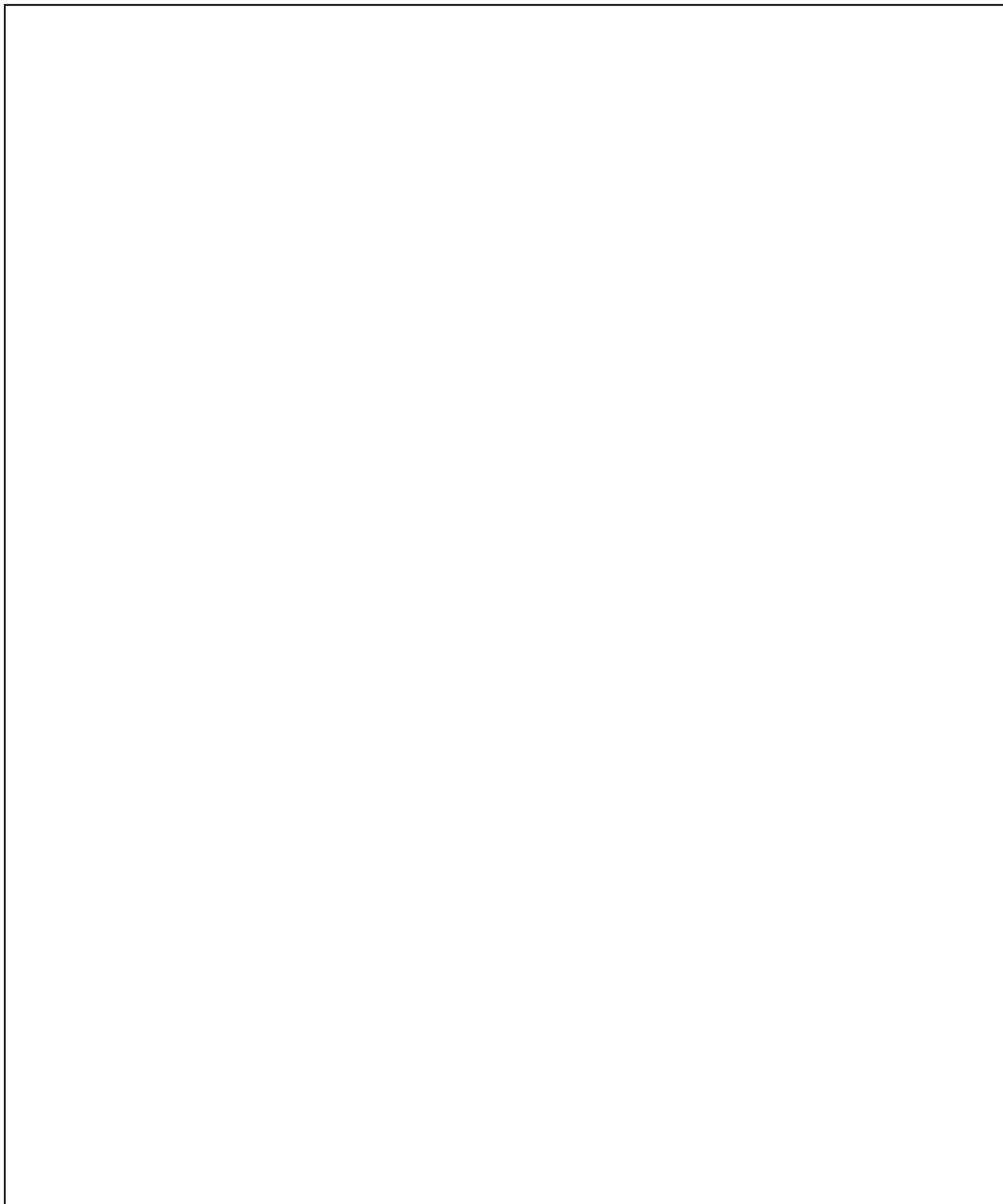
7-4 Internal Coaxial Cable Replacement

- 7-4-1 Remove the external coaxial cable as described in Section 7-2.
- 7-4-2 Remove the retaining nut and washer from the SMA feedthrough connector in the bottom cover using a 3/8 inch open end wrench. See ILLUSTRATION 7-3.
- 7-4-3 Disassemble the unit as described in Section 7-3.
- 7-4-4 Remove the binding posts, items 1 and 2 in the illustration detail, using a flat blade screwdriver.
- 7-4-5 Remove the cable clamp, item 3, using a flat blade screwdriver and a 5/16 inch open end wrench.
- 7-4-6 Remove the SMA feedthrough connector from the bottom cover.
- 7-4-7 Withdraw the internal coaxial cable from the cable management system. See ILLUSTRATION 7-5.
- 7-4-8 Feed the new cable in the same path as the old. Push the SMA feedthrough connector through the hole in the bottom cover and secure with the nut and washer to 10 inch-pounds.
- 7-4-9 Place the cable clamp (item 3) on the cable 5 1/4 inches from the SMA fitting. Orient the clamp as shown in ILLUSTRATION 7-5. Tighten to 3 inch-pounds.
- 7-4-10 Install new binding posts and screws (items 1 and 2) to reconnect the cable management system to the Coil assembly and the mounting bracket. Push the parts together. Do not thread the screw into the post.

NOTE

The cable management linkage bars must be attached to the bottom side of the Coil assembly and the mounting bracket.

- 7-4-11 Verify the operation of the drive system by sliding the connector block several times from one end of its travel to the other.
- 7-4-12 Assemble the coil housing following the steps in Section 7-6.
- 7-4-13 Repeat the tests in Sections 3-0, 4-0, and 5-0 to verify proper coil operation.



DRIVE CORD ADJUSTMENT AND REPLACEMENT

ILLUSTRATION 7-6

7-5 Drive Cord Adjustment and Replacement

- 7-5-1 Disassemble the unit as described in Section 7-3.
- 7-5-2 To tighten the drive cord, loosen the lock nuts [items 4] and tighten the tension adjuster bolts [items 5] to obtain alignment between the coil and the indicator. Tighten the lock nuts one-quarter turn beyond contact with the indicator block. See ILLUSTRATION 7-5
- 7-5-3 To replace the drive cord, remove the lock nuts [items 4] and the tension adjuster bolts [items 5]. Remove the cord from the coil assembly.
- 7-5-4 Place the new cord in the appropriate holes in the coil assembly from the top. Tie a double knot in the cord underneath the coil assembly and cut off the excess cord. See ILLUSTRATIONS 7-5, 7-6.
- 7-5-5 Route the cord around the inner posts closest to the coil assembly and crisscross the two cords. Route the cord around the outer posts on the opposite sides of the coil housing.
- 7-5-6 Feed the opposite end of the cord through the tension adjuster bolts and tie a double knot to achieve the required finish length. Cut off the excess cord.
- 7-5-7 Replace the tension adjuster bolts in the indicator block. Adjust the bolts as described in step 7-5-2.
- 7-5-8 Verify the operation of the drive system as described in step 7-4-11.
- 7-5-9 Assemble the coil housing as described in Section 7-6.
- 7-5-10 Repeat the tests in Sections 3-0, 4-0, and 5-0 to verify proper coil operation.

7-6 Assembly of the Coil and Positioner - ILLUSTRATION 7-4

- 7-6-1 With the coil assembly set to the center of its travel, place the top cover on the bottom cover. Spread the slot in the top cover sufficiently to pull the positioning handle through. Press the top cover down on the bottom cover, making certain the ten brass studs protrude through the holes in the bottom cover.
- 7-6-2 While holding the top and bottom covers securely together, turn the coil over on the top cover. Install the ten brass washers and nuts on the ten studs. Torque to 20 inch-pounds. Slide the coil from one end of travel to the other to ensure that there is no binding of the coil positioning hardware.
- 7-6-3 Insert nine new drive rivets supplied with the service kit into the appropriate holes in the positioner housing. Using a small hammer, drive the center pins into the rivet bodies until flush.
- 7-6-4 Attach the carrying handle using the two knurled nuts and two springs.
- 7-6-5 Replace the nine nylon screws located beneath the scale. Torque to 2 inch-pounds.
- 7-6-6 Install the external cable as described in Section 7-2.
- 7-6-7 Connect the internal coaxial cable to the SMA fitting on the balun board as noted in step 7-3-3.

7-6-8 Attach the clear service access cover using the six nylon screws. Torque to 1 inch-pound.

7-6-9 Return the coil housing to its normal position and attach the comfort pad.

7-6-10 Repeat the tests in Sections 3-0, 4-0, and 5-0 to verify proper coil operation.

7-7 Output Connector Replacement ILLUSTRATION 7-7

7-7-1 If the cable will be shortened to less than 71 inches by a repair, it is recommended that the entire cable be replaced in the event of a connector or connector-end cable failure.

7-7-2 Replace the connector using ILLUSTRATION 7-7 as a guide. Use a new Tri-Lug BNC coaxial fitting - part number 46-317498P1, and crimp tool - part number 46-255841, with insert - part number 46-255841P100, to install the replacement connector on the cable.

7-7-3 Repeat the tests in Sections 3-0, 4-0, and 5-0 to verify proper coil operation.

7-8 Diode Replacement

7-8-1 As coil tuning, orthogonality, and decoupling may be affected by decoupling diode replacement, decoupling diodes are not field replaceable. Coils with decoupling diode failure must be replaced.

7-8-2 The protection diodes are in parallel. One of the diodes is inaccessible and cannot be changed. Coils with protection diode failure must be replaced.

OUTPUT CONNECTOR REPLACEMENT

ILLUSTRATION 7-7

DEFECTIVE SURFACE COIL RETURN FORM

NOTE

To allow for proper assessment of defective returned coils, this form must be completely filled out and accompany all returned coils. Include films or prints of any image quality complaints with a description of the scan prescription used.

DATE: _____

SITE NAME: _____

SITE ADDRESS: _____

SERVICE ENGINEER: _____

COIL SERIAL NUMBER: _____

DATE COIL INSTALLED: _____

DESCRIPTION OF COIL PROBLEM:

ELECTRICAL CHECKS	
VECTOR IMPEDANCE METER TEST - COIL LOADED WITH HUMAN SUBJECT IN FREE SPACE	
MAGNITUDE AT 21.29 MHz	[50 OHMS]
PHASE AT 21.29 MHz	[0°]
VECTOR IMPEDANCE METER TEST - COIL UNLOADED IN FREE SPACE	
MAGNITUDE AT 21.29 MHz	
PHASE AT 21.29 MHz	